

Answer Key — Biology Grade 11

Sub-Strand 1.1: Taxonomy I • Lesson 7: Five Kingdoms of Living Organisms

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	B	C	C	B	A	B

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share on how scientists would group the eight organisms

Q1. Groups once sorted a goat, chicken, maize, mushroom, algae, bacteria, bean plant and mosquito by colour, habitat and usefulness, and every group was right by its own rules. Which basis for grouping can be checked by anyone, in any country, and give the same result?

- A. The colour and size of the organism
- B. How useful or harmful the organism is to people
- C. Cell type and how the organism gets its food**
- D. What the organism is called in Kikuyu, Luo or Kiswahili

Correct answer: C. Cell type and mode of nutrition are structural and feeding characteristics that can be checked by anyone. Colour, size, usefulness and local names change from person to person or place to place.

USE AFTER: OBSERVE PHASE — Jigsaw Step 1: expert groups completing the Kingdom Data Card (Monera evidence pack)

Q2. Learners studying bacteria and the fermented milk maziwa lala record the cell type on their Kingdom Data Card. Which kingdom is made of prokaryotic cells?

- A. Plantae
- B. Monera**
- C. Protocista
- D. Fungi

Correct answer: B. Monera (bacteria and cyanobacteria) is the only kingdom of prokaryotic cells with no true nucleus. Protocista, Fungi, Plantae and Animalia are all eukaryotic.

USE AFTER: OBSERVE PHASE — Jigsaw Step 1: Protocista expert group (Spirogyra or green pond water, algae from Lake Victoria)

Q3. A learner looks at green Spirogyra from a pond jar and says, 'It is green and lives in water, so it is a plant.' Which is the best response?

- A. Correct, all green organisms belong to Kingdom Plantae
- B. Wrong, Spirogyra is an animal because it lives in water
- C. Wrong, Spirogyra and other algae belong to Kingdom Protocista, so being green does not make an organism a plant**
- D. Wrong, Spirogyra is a fungus because it grows in water

Correct answer: C. Algae such as Spirogyra are placed in Protocista. Kingdoms are decided by shared characteristics such as cell type, organisation and nutrition, not by colour or habitat.

USE AFTER: EXPLAIN PHASE — Jigsaw Step 2: Claim-Evidence-Reasoning debate on the mushroom

Q4. A mushroom grows from the soil next to a maize plant. Why is the mushroom NOT placed in Kingdom Plantae?

- A. It is a prokaryote, so it cannot be a plant
- B. It is not green, so it must be an animal
- C. It cannot make its own food (it is heterotrophic) and its cell wall is made of chitin**
- D. It grows in soil, which only fungi do

Correct answer: C. A mushroom is a fungus. It feeds on organic matter instead of making food by photosynthesis, and its cell wall contains chitin. Growing in soil does not make an organism a plant.

USE AFTER: EXPLAIN PHASE — Jigsaw Step 2: Claim-Evidence-Reasoning and card sort of the eight organisms

Q5. A learner writes a Claim-Evidence-Reasoning statement placing the mosquito in Kingdom Animalia. Which reasoning is the strongest?

- A. It is a pest that spreads disease
- B. It is a multicellular eukaryote that feeds on other organisms (heterotrophic)**
- C. It lives near water and is active at night
- D. It is small and has wings, like a chicken

Correct answer: B. Animalia is defined by cell type, organisation and nutrition: multicellular, eukaryotic and heterotrophic. Being a pest, having wings or living near water are appearance, habitat or usefulness ideas, not kingdom characteristics.

USE AFTER: MODEL BUILDING PHASE — Model: groups revise the class classification model with the eight phenomenon organisms placed in kingdoms

Q6. A group revises its class model and places the eight organisms into kingdoms. Which placement is correct?

- A. Bacteria in Monera, algae in Protocista and mushroom in Fungi**
- B. Bacteria in Protocista and goat in Animalia
- C. Algae in Plantae and mushroom in Fungi
- D. Mushroom in Plantae and bacteria in Monera

Correct answer: A. Bacteria are prokaryotes (Monera), algae are in Protocista, and the mushroom is a fungus. Placing algae or the mushroom in Plantae, or bacteria in Protocista, is a common error.

USE AFTER: END OF LESSON — End of lesson: summary of the five-kingdom system

Q7. Which list names the five taxonomic kingdoms of living organisms?

- A. Monera, Fungi, Plantae, Animalia, Viruses
- B. Monera, Protocista, Fungi, Plantae, Animalia**
- C. Plantae, Animalia, Fungi, Bacteria, Insects
- D. Bacteria, Algae, Mushrooms, Plants, Mammals

Correct answer: B. The five kingdoms are Monera, Protocista, Fungi, Plantae and Animalia. The other lists include groups such as algae, mammals, insects or viruses that are not kingdoms in this system.